



FACULTY OF INFORMATION TECHNOLOGY
BACHELOR OF SCIENCE IN INFORMATICS AND COMPUTER SCIENCE
END OF SEMESTER EXAMINATION
ICS 1202 OBJECT ORIENTED PROGRAMMING

DATE: 23rd November 2018

Time: 2HRS

Instructions

1. This examination consists of **FIVE** questions.
2. Answer **Question ONE (COMPULSORY)** and any other **TWO** questions.

Question One (30 marks)

- a) Object oriented programming was introduced as a programming paradigm to solve the challenges experienced in procedural programming. Name the four possible challenges faced when using procedural programming paradigm. **(4mks)**
- b) Using an example object in real life, describe the three components of an object as outlined in object-oriented programming. **(3mks)**
- c) Constructors are important components of a Class in OOP. Explain their main function,
then use Java to write code for a simple constructor demonstrating its use. **(4mks)**
- d) Working with JDBC in java involves a process of six steps. Describe each step in detail **(9mks)**
- e) Create a Java program containing a class that simulates a Bank's functionalities of creating a bank account, making a deposit, and withdrawing money from the account. This program should make use of constructor(s), methods, and any other OOP concepts learnt. **(10mks)**

Question Two (15 marks)

- a. Consider the following class definition that represents a thermostatically controlled home heating system.

```
class thermostaticHeaterSystem
{
    public:
        int temperatureLimit; // from 0 to 30 (celsius)
        bool heaterState; // true or false (for on/off)
        void evaluateState(); // switch heater on/off as needed
        int getCurrentTemp(); // obtain temperature from sensor
        bool getCurrentHeaterState(); // report whether on or off
};
```

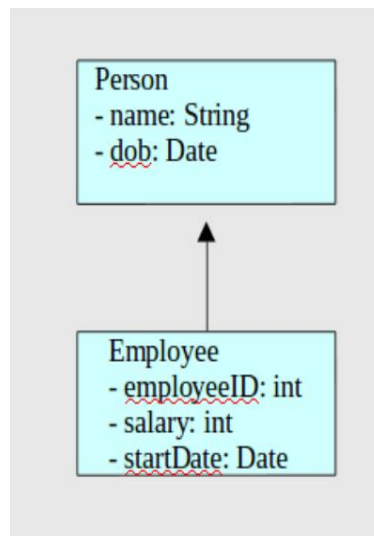
- i. Provide a redesigned thermostatic Heater System class that uses more appropriate access modifiers. **(5 marks)**
 - ii. Provide a getter and a setter method that will enable the temperatureLimit instance variable to be retrieved and modified. **(3 marks)**
 - iii. Provide a constructor for the ThermostaticHeaterSystem class that will initialise the instance variables to suitable (valid) start values. **(3 marks)**
- b. List and explain four types of data types that are primitive to Java Programming Language **(4mks)**

Question Three (15 marks)

- a. Access modifiers can be used to specify the visibility of a class and its members. Define the three access modifiers used in Java **(6mks)**
- b. Exceptions are bound to happen every time we execute a java application.
- i. Describe the two types of exceptions in Java. For each type, provide an example. **(4mks)**
 - ii. Using Java code snippets, create a class that contains a try catch block statement to illustrate an InputMismatchException. The program should ask the user to input an integer and while handling the exception, it should display a message informing the user that a number format exception has occurred. **(5mks)**

Question Four (15 marks)

- a. An abstract class is a class that contains one or more abstract methods.
- Using Java code snippet, define an abstract class `LivingThing` with two methods `breath` and `eat` and an abstract method `walk`. The methods should enable the user to print an output of the actions on a console window E.g. *Living thing eating, living thing breathing* **(4mks)**
 - Describe another class `human` that implements the abstract method `walk` defined in the above section. The method should enable the user to print an output the actions on a console window e.g. *Human walks* **(2mks)**
 - Describe a second class `donkey` that implements the abstract method `walk` defined in the living thing class **(2mks)**
- b. Inheritance is a significant property that the OOP paradigm has among its other properties. The figure below represents a simple UML diagram representation of inheritance. Write the equivalent java code for the UML diagram to illustrate the concept of inheritance. **(7mks)**



Question Five (15 marks)

- a. Differentiate between overloading methods and overriding methods as used in Java Programming. (4mks)

- b. List down the four rules for overridden methods in java (4mks)
- c. Write a swing class that will produce the following output (7mks)

